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Editors
Journal of the American Statistical Association

Dear Editors,

re: “Quantifying Moderator Effects on High-Dimensional Multi-Subject Time Series” by José Á. Sánchez Gómez, Holly O’Rourke and Gene Brewer.

Could you please consider the above article for possible publication in *JASA*?

Researchers in many scientific fields use time series data to estimate the causal effects that variables have on each other over time. These relationships can be modeled via Granger-causal networks (GCN), where nodes correspond to variables, and directed links indicate lagged temporal relationships among them. When time series data is gathered over multiple subjects, GCN models must account for potential subject heterogeneity. While methods in the literature can perform estimation of heterogeneous GCNs, they often lack interpretable explanations of the source of such measured heterogeneity. In the present paper, we propose the moderator vector auto-regressive model (MOD-VAR) for interpretable estimation of heterogeneous GCN in multi-subject time series data. We achieve this by using external moderator variables to explain the heterogeneity of GCNs across subjects. Our theoretical results confirm the improved estimation of MOD-VAR for heterogeneous GCN estimation in high dimensional scenarios, even when the GCN are highly heterogeneous. Furthermore, our extensive numerical simulations confirm the superior performance of our proposed MOD-VAR for GCN estimation and forecasting in a wide variety of settings. We analyze resting state fMRI data with time-invariant moderators and showcase the usefulness of recovering moderating effects on the estimation of GCNs for multi-subject data.

Thank you very much for your consideration. I look forward to hearing from you.

Yours sincerely

José Á. Sánchez Gómez